

Week 4 (Thursday, July 17)

Overfitting, Regularization, Validation

CMPE-257: Machine Learning - Summer 2025



SJSU

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Today's plan...

Main Topic: Overfitting, Regularization, Validation

5:30-6:40 Session 1

6:40-7:10 Break

7:10-9:00 Session 2

9:00-9:30 Term Projects and HW4

Session 1:

- Overfitting and Regularization

Session 2:

- Cross-validation
- Hands-on coding session 4

Third Homework is Due on Sunday!

Week 4 - Session 1

Overfitting and Regularization

- Overfitting
- Subset Selection
- Regularization



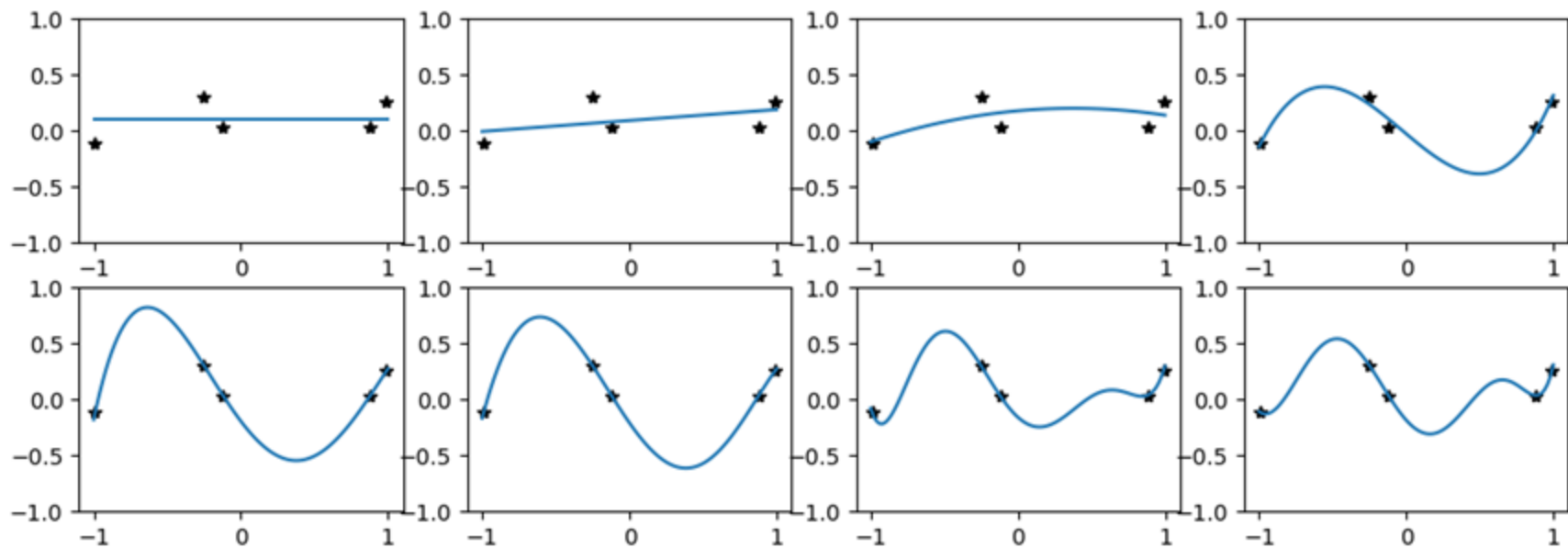
The learning problem revisited

$$\mathbf{E}_{\text{out}}(\mathbf{g}^*) \leq \mathbf{E}_{\text{in}}(\mathbf{g}^*) + \text{Generalization Bound}$$

- Make the in-sample error $\mathbf{E}_{\text{in}}(\mathbf{g}^*)$ small enough
- Ensure the out-of-sample error $\mathbf{E}_{\text{out}}(\mathbf{g}^*)$ is close enough to the in-sample error $\mathbf{E}_{\text{in}}(\mathbf{g}^*)$
 - i.e. make the Generalization Bound as small as possible

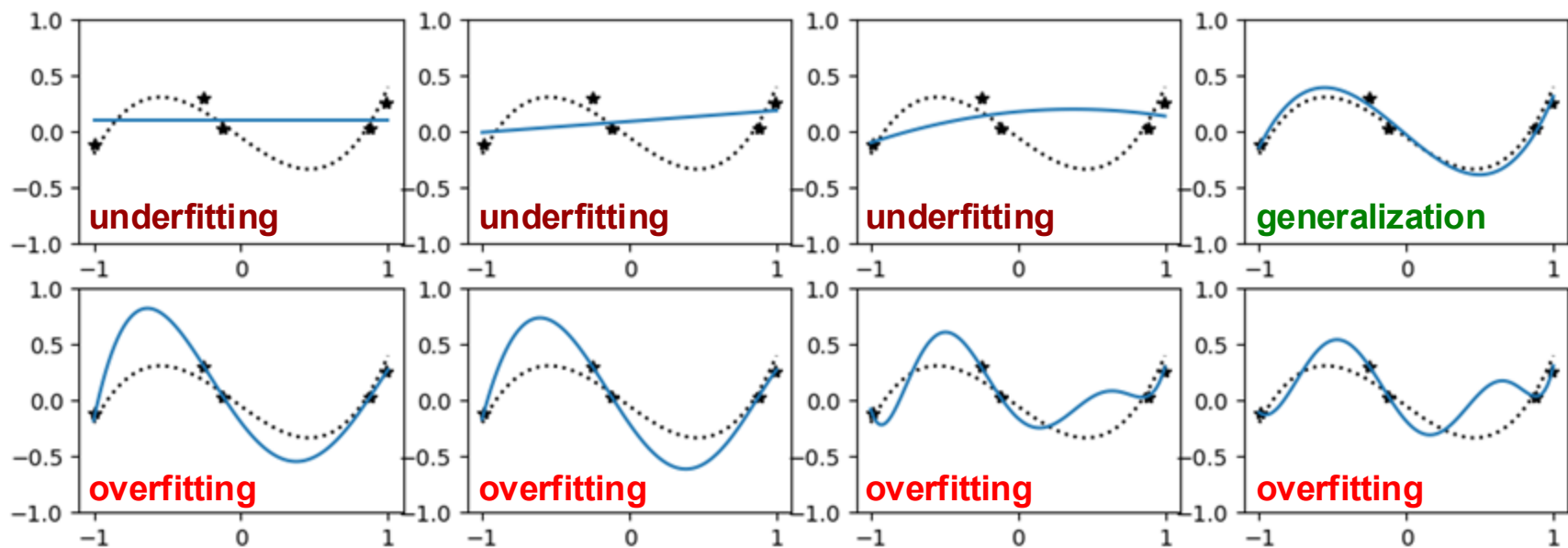
Attempting to learn from five data points

- Which of the models is actually learning?



Attempting to learn from five data points

- We need to look at out-of-sample errors to make the assessment

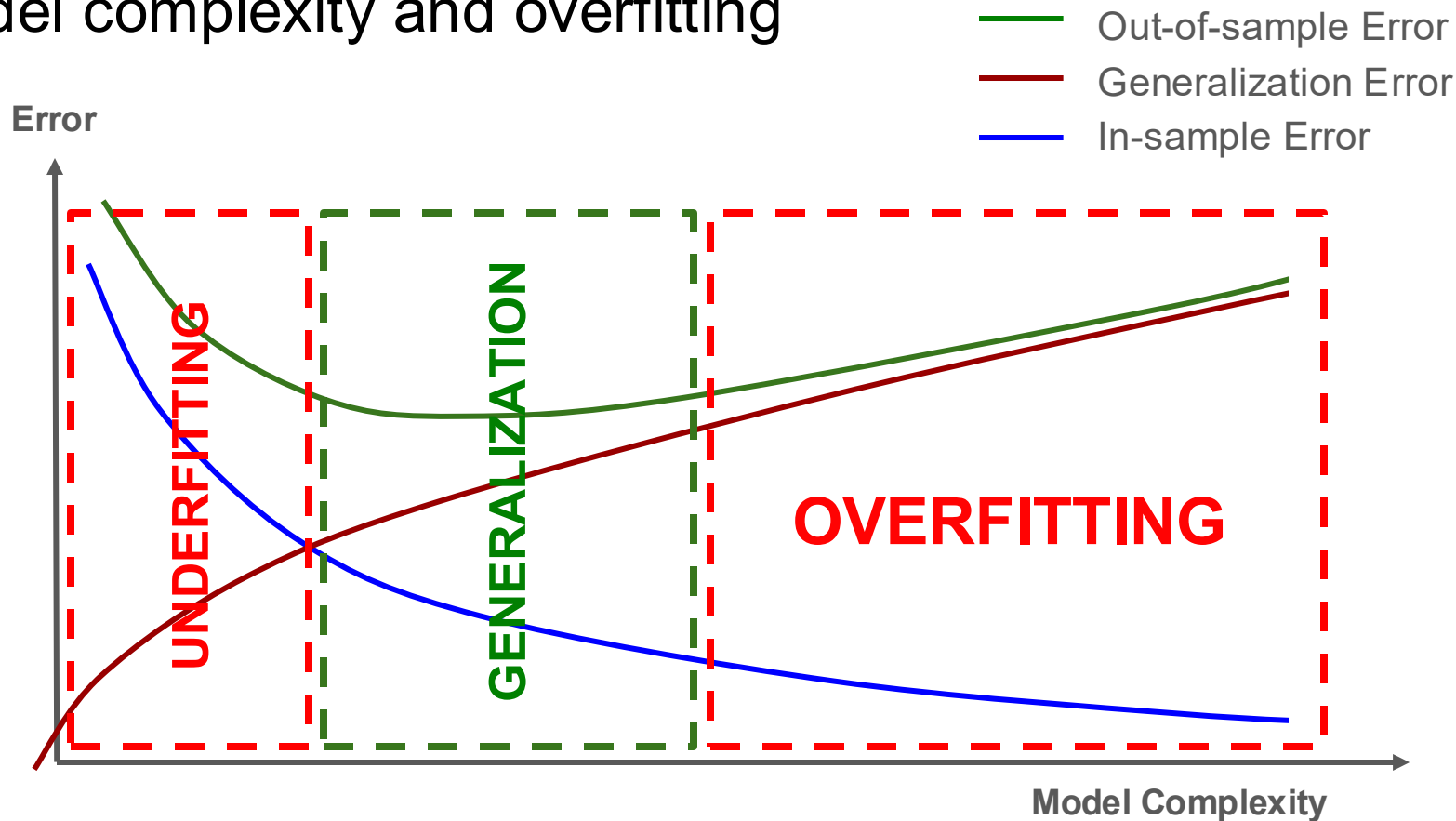




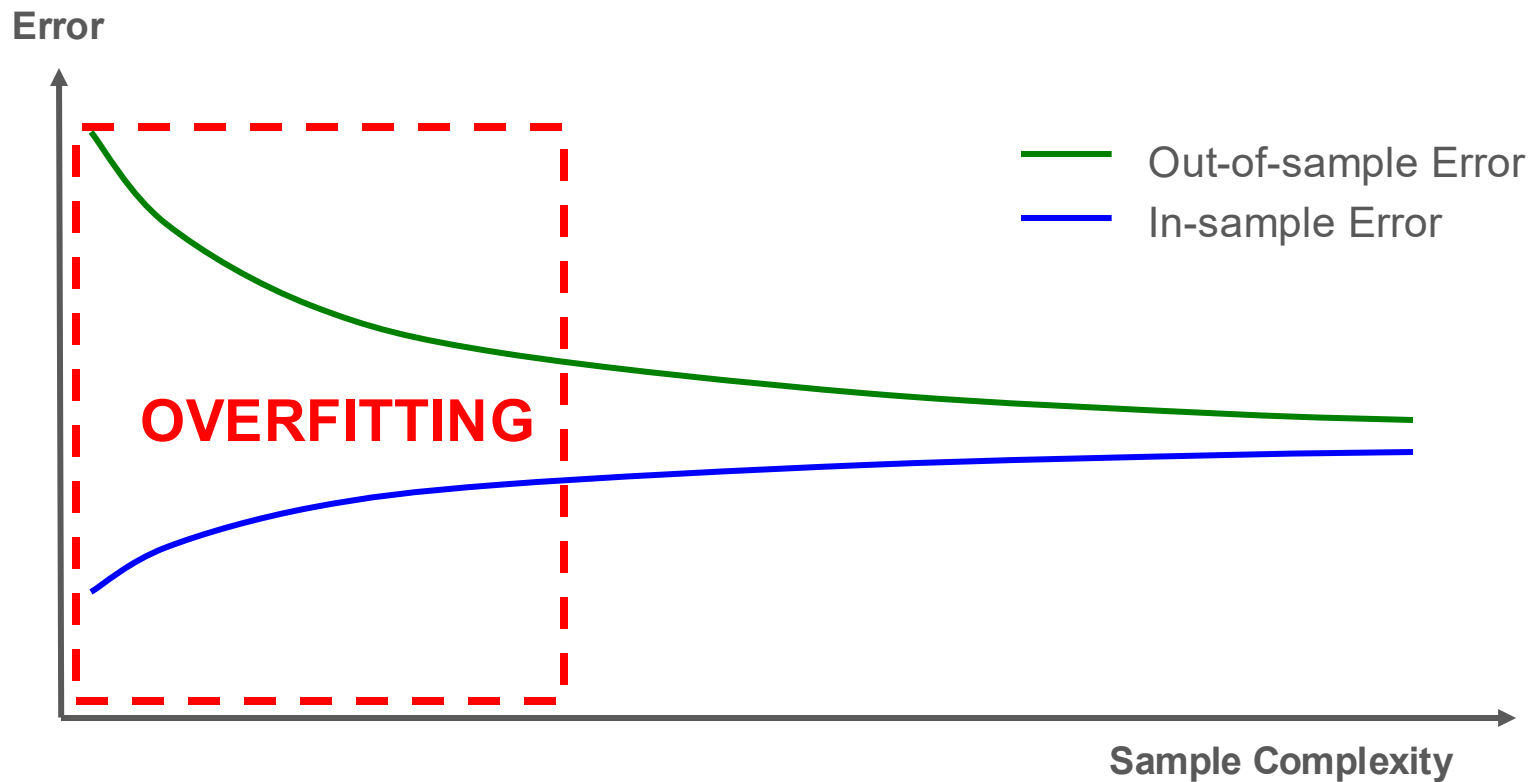
Underfitting, overfitting, and generalization

- Underfitting: the model cannot fit the data
 - Large in-sample and out-of-sample errors
- Overfitting: the model fits the data more than needed
 - Small in-sample error and large out-of-sample error
- Generalization: the model fits the data to the right extent
 - Small in-sample and out-of-sample errors

Model complexity and overfitting



Sample complexity and overfitting



Two fundamental questions...

Why the model overfits the data rather than learning the right complexity (i.e. $y = a + b x + c x^2 + d x^3 + \mathbf{0} x^4 + \mathbf{0} x^5 + \dots$) ?

- Overfitting results from interactions of three factors: stochastic noise, deterministic noise, and number of in-sample data points

How we overcome the overfitting problem in practice?

- Either reducing or penalizing the degrees of freedom of the model: subset selection and regularization

Deterministic noise and stochastic noise

- Deterministic noise: the part of the target function that is not properly fit by the model
- Stochastic noise: the inherent noise ξ that comes with the data

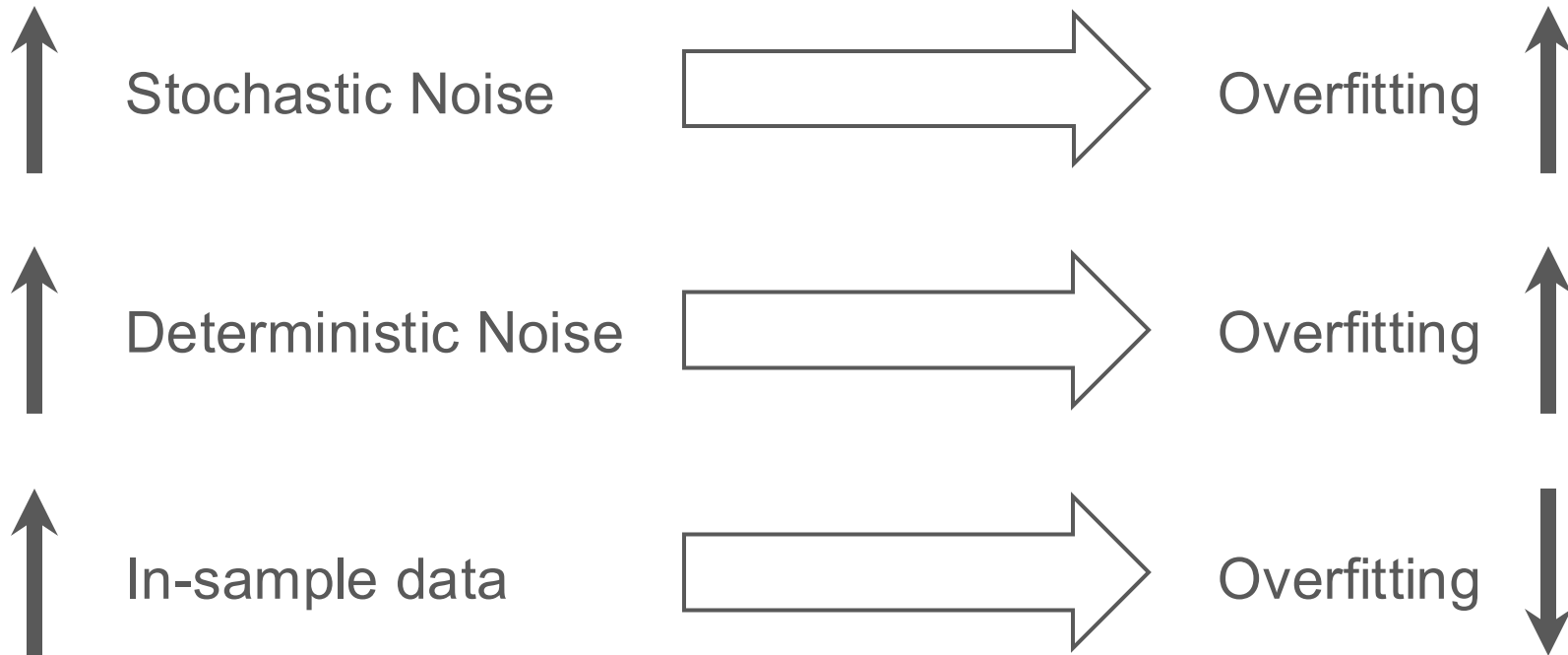
$$\underbrace{E[(f(x) + \xi - g(x))^2]}_{\text{out-of-sample MSE}} = \underbrace{(f(x) - \bar{g}(x))^2}_{\text{BIAS}^*} + \underbrace{E[(g(x) - \bar{g}(x))^2]}_{\text{VAR}} + \underbrace{\text{var}(\xi)}_{\text{Irreducible}}$$

Deterministic Noise $\xrightarrow{\text{dashed arrow}}$ **BIAS***

Stochastic Noise $\xrightarrow{\text{dashed arrow}}$ **VAR**

Stochastic Noise $\xrightarrow{\text{solid arrow}}$ **Irreducible**

Overfitting vs noise and data

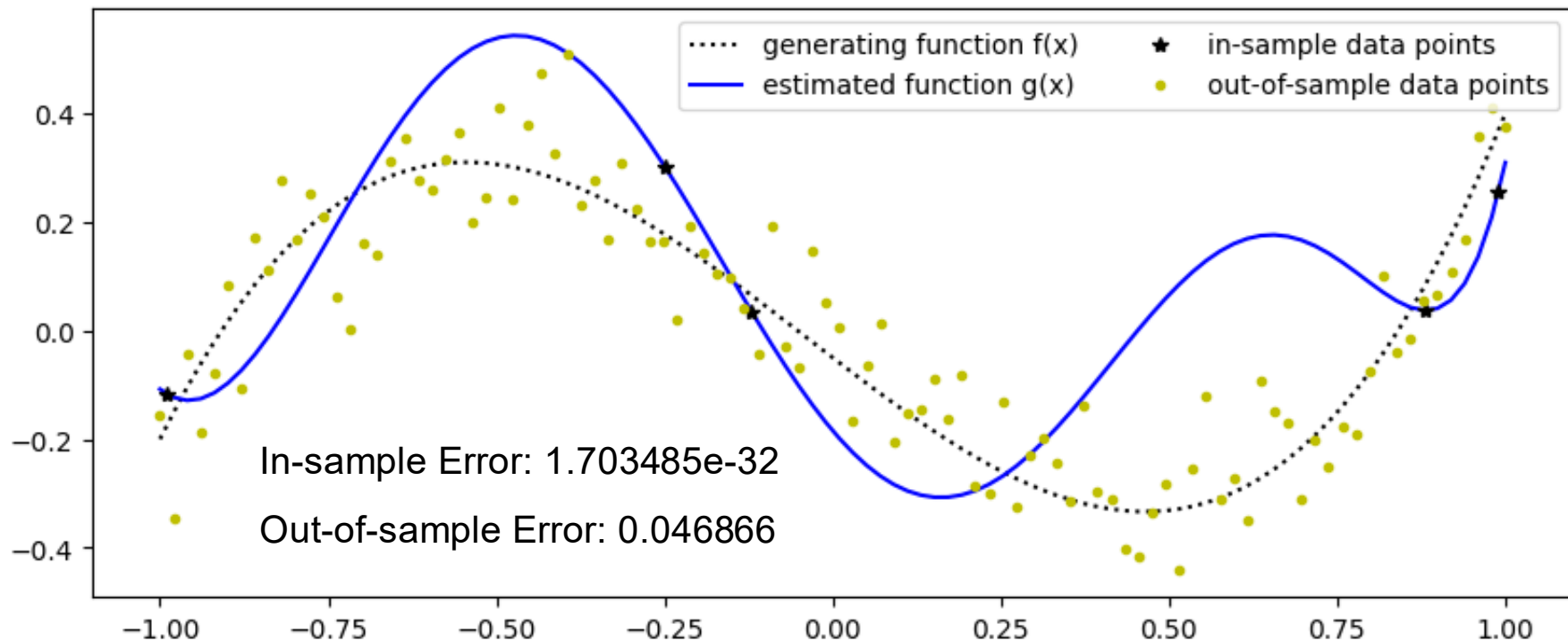


Overcoming overfitting via subset selection

Identify the subset of predictors that better explains the target

- Best subset selection: fit all possible models by considering all possible predictor combinations and select the best performer
- Stepwise selection: explore a restricted set of models
 - Forward stepwise selection: add one predictor at a time
 - Backward stepwise selection: remove one mode at a time
 - Hybrid: combines forward and backward selections
- Dimensionality reduction: smaller set of transformed predictors

Consider a baseline regression model (with 8 predictors)



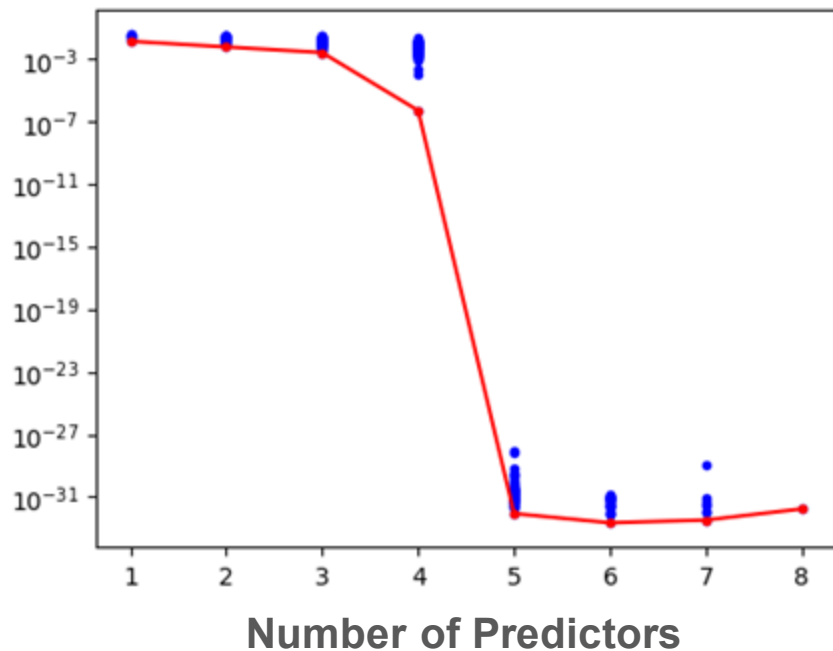
Let us conduct best subset selection

$$y = w_0 p_0 + w_1 p_1 + w_2 p_2 + w_3 p_3 + w_4 p_4 + w_5 p_5 + w_6 p_6 + w_7 p_7 + \xi$$

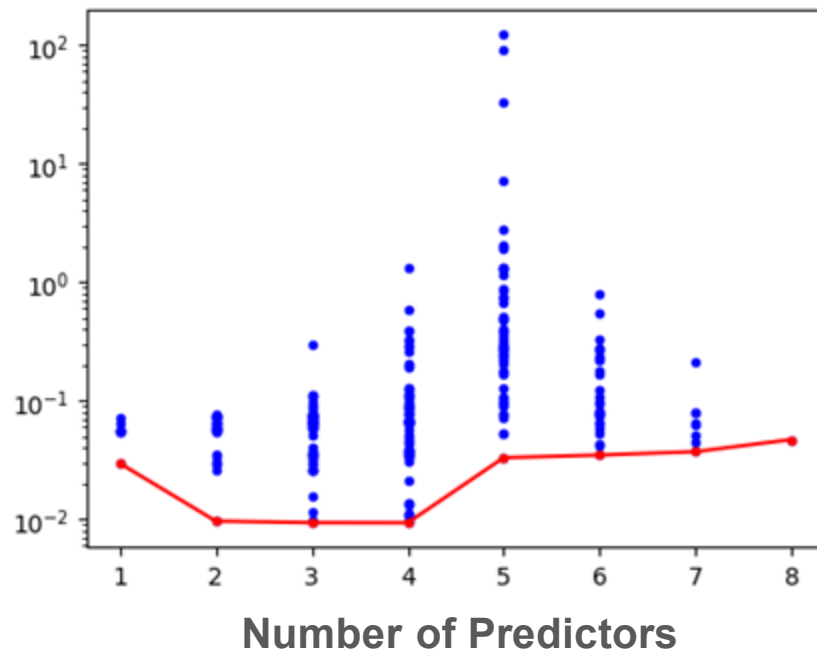
- Identify the subset of predictors $\{p_k\}$ that better explain y
- Estimate the best regression coefficients for such subset
- Best subset selection procedure:
 - Train all possible models: $\sum_n C(8, n) = 256$
 - Retain the model with lowest out-of-sample error

Performance of all possible models

In-sample Errors



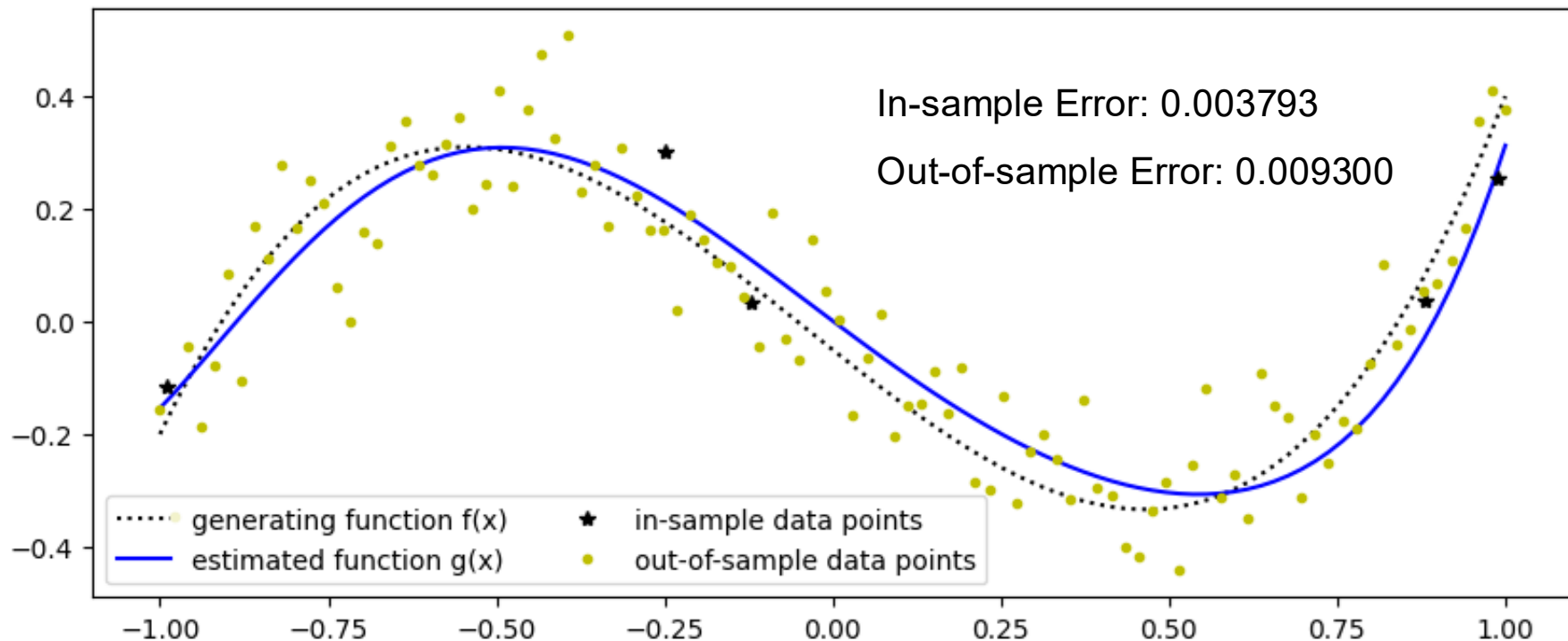
Out-of-sample Errors



Best subset selection

# Predictors	In-sample Error	Out-of-sample Error	Selected Subset
2	5.613881e-03	0.009612	$\{p_1, p_3\}$
3	3.927155e-03	0.009329	$\{p_1, p_3, p_6\}$
4	3.793522e-03	0.009300	$\{p_1, p_3, p_4, p_6\}$
5	4.068527e-32	0.032965	$\{p_1, p_3, p_4, p_5, p_6\}$

Optimal regression model with $\{p_1, p_3, p_4, p_6\}$



Subset selection via dimensionality reduction

- An alternative approach to subset selection is to transform the predictors $\{p_k\}$ into a smaller subset of “better” predictors $\{z_m\}$

$$z_m = \sum_{k=1}^K \phi_{km} p_k \quad \text{for } m = 1, 2, \dots, M \quad \text{with } M \ll K$$

- Then, perform the regression against the resulting subset of transformed predictors $\{z_m\}$
- The transformation is typically designed to reduce noise and concentrate information within a smaller dimensionality space



Singular value decomposition (SVD)

- A factorization of a real-valued matrix $\mathbf{M}$ ($n \times k$) into a diagonal matrix $\mathbf{W}$ ($n \times k$) and orthonormal matrices $\mathbf{U}$ ($n \times n$) and $\mathbf{V}$ ($k \times k$)

$$\mathbf{M} = \mathbf{U} \mathbf{W} \mathbf{V}^T$$

- $\sigma_i = \mathbf{W}_{ii}$ are referred to as the singular values of $\mathbf{M}$
- The columns of $\mathbf{U}$ and $\mathbf{V}$ are the left-singular and right-singular vectors of $\mathbf{M}$, i.e. $\mathbf{M}^T \mathbf{u}_i = \sigma_i \mathbf{v}_i$ and $\mathbf{M} \mathbf{v}_i = \sigma_i \mathbf{u}_i$
- Notice SVD is not a dimensionality reduction technique!



Principal component analysis (PCA)

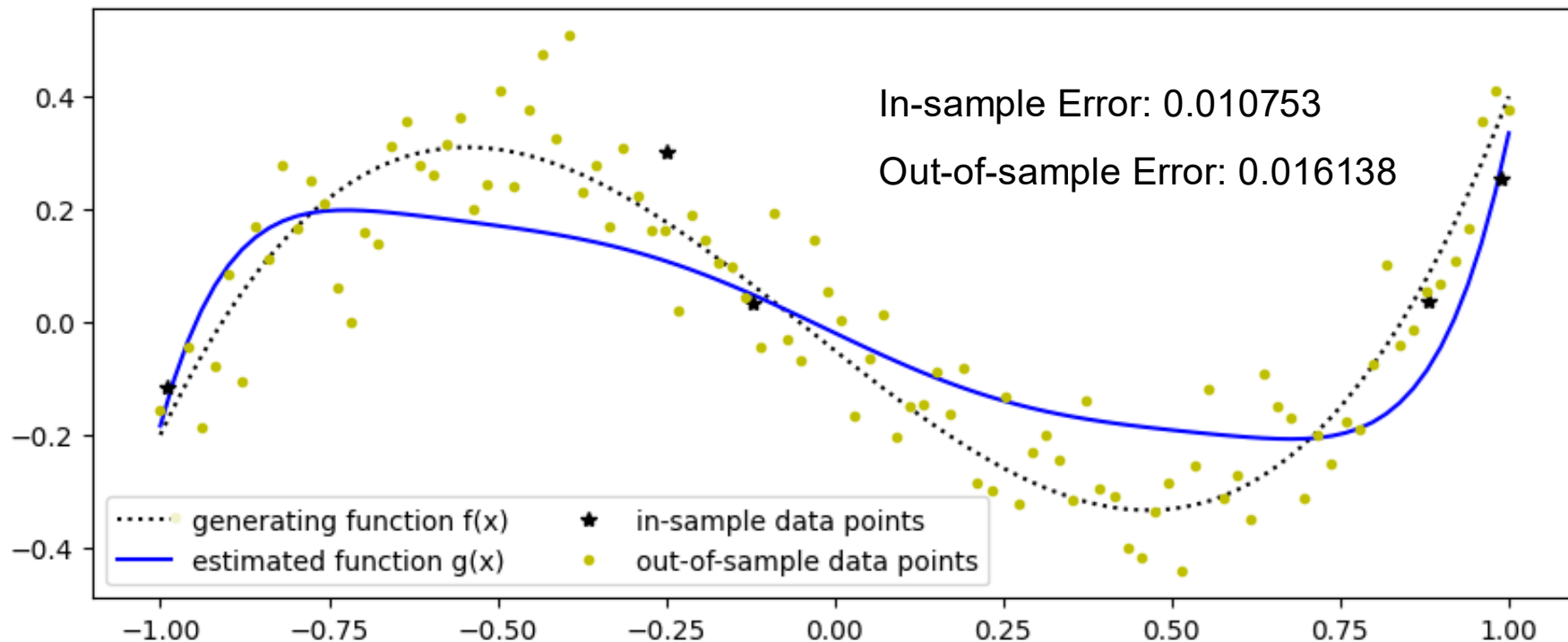
- Data **M** is transformed into **T** such that directions exhibiting largest variances (principal components) are identified
- The principal components constitute an orthonormal basis

$$\mathbf{T} = \mathbf{M} \mathbf{V} = \mathbf{U} \mathbf{W} \quad \text{where} \quad \mathbf{M} = \mathbf{U} \mathbf{W} \mathbf{V}^T$$

- Dimensionality reduction is achieved by retaining the first S largest singular values σ_i where $S \ll K$

$$\mathbf{T}_{(n \times s)} = \mathbf{M}_{(n \times s)} \mathbf{V}_{(s \times s)} = \mathbf{U}_{(n \times n)} \mathbf{W}_{(n \times s)}$$

Regression model with PCA ($S = 3$)



Overcoming overfitting via regularization (shrinkage)

- Another way of overcoming overfitting is by restricting or penalizing the ability of the model to fit the in-sample dataset

$$\text{MSE}_{\text{aug}} = \frac{1}{N} \sum_{n=1}^N (y_n - \hat{y}_n)^2 + \text{penalty}$$

- The penalty term is used to constraint the solution space for the coefficients (typically by limiting their size)



Ridge (L2) regularization

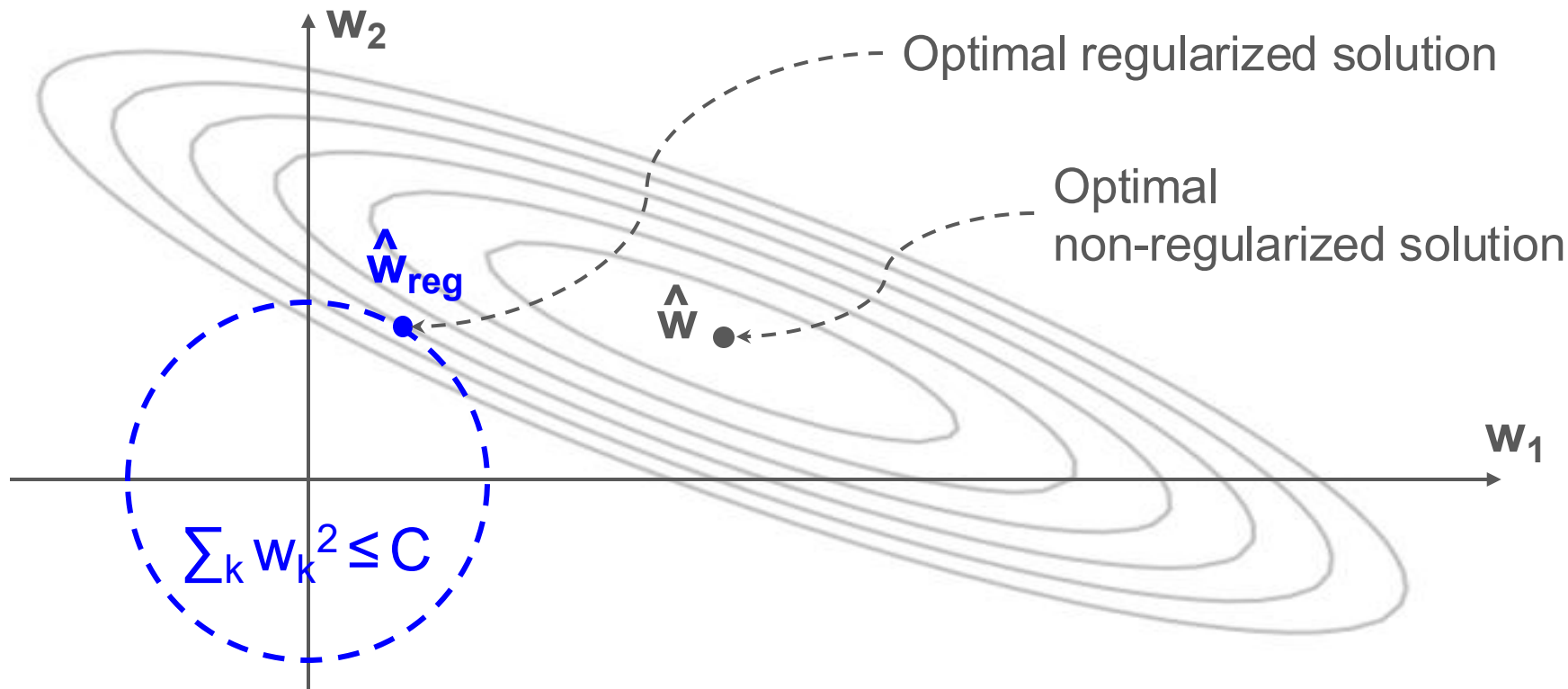
- Minimizes the following augmented error score:

$$\text{MSE}_{L_2}(\mathbf{w}) = \text{MSE}(\mathbf{w}) + \lambda \sum_k w_k^2$$

- Where $\lambda > 0$ controls the relative impact of the penalty
 - Large λ : the coefficients w_k are forced to be small
- Minimizing augmented error score MSE_{L_2} is equivalent to solving the constraint minimization problem:

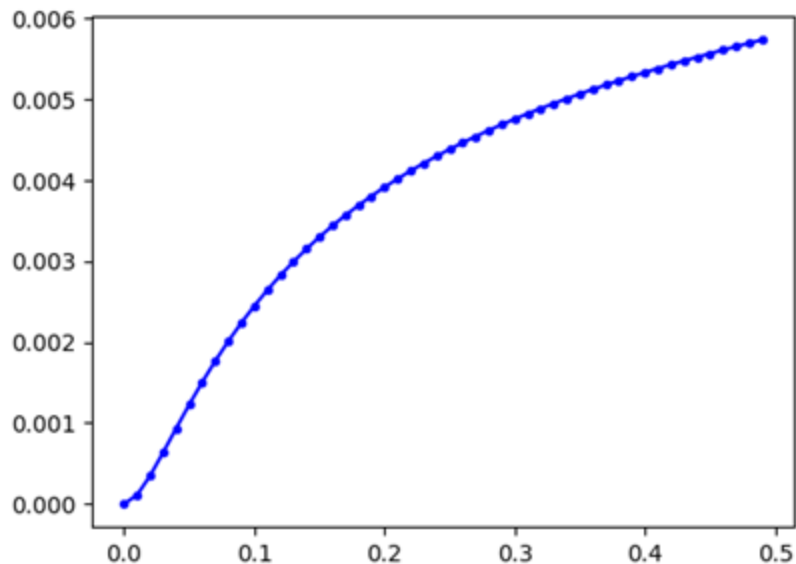
$$\text{argmin } \text{MSE}(\mathbf{w}) \text{ subject to } \sum_k w_k^2 \leq C$$

Understanding the role of the constraint region (Ridge)



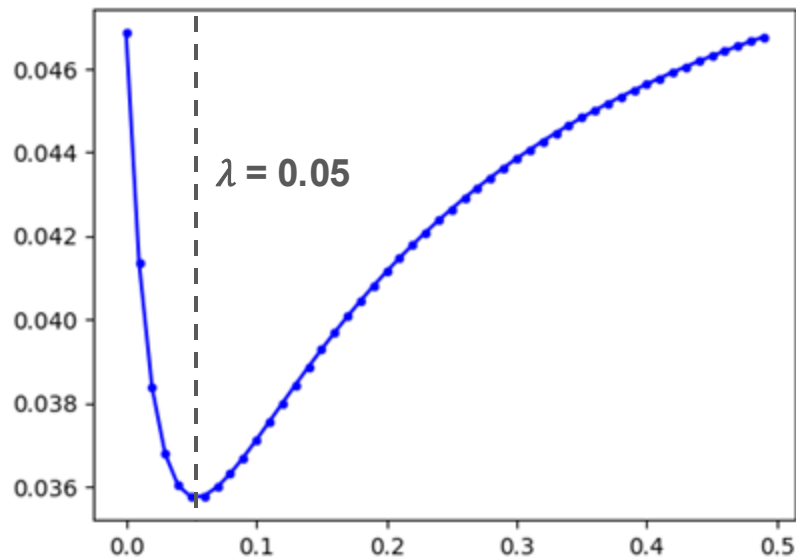
In- and out-of-sample errors as functions of λ (Ridge)

In-sample Errors



Shrinkage factor λ

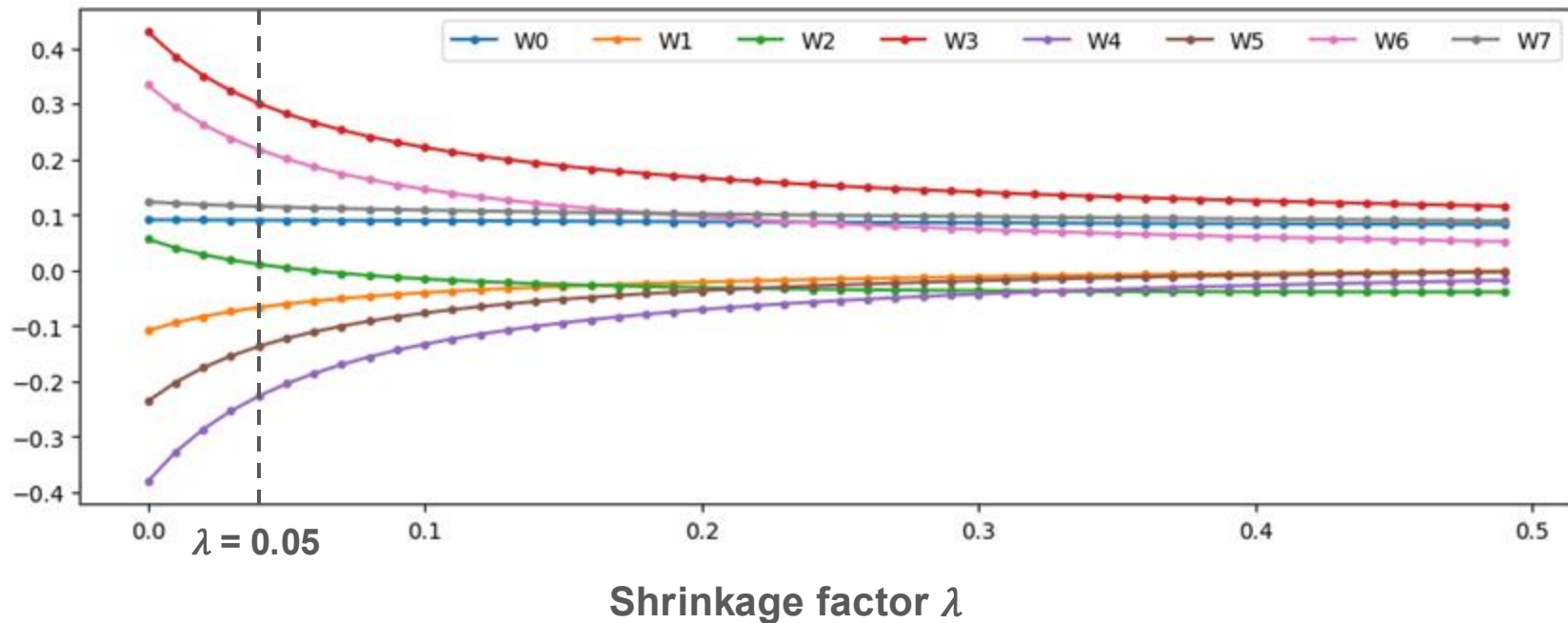
Out-of-sample Errors



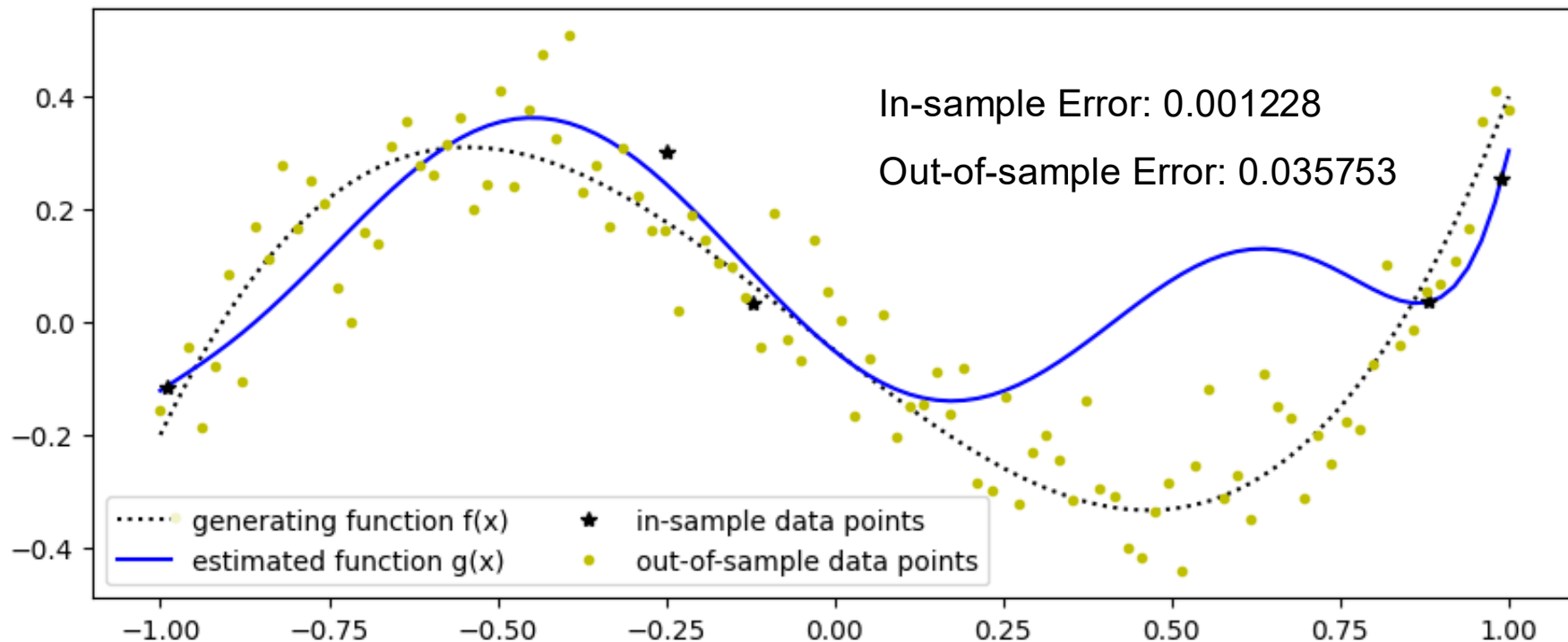
Shrinkage factor λ

Optimal model coefficients as functions of λ (Ridge)

Optimal model coefficients



Ridge regression model (with $\lambda = 0.05$)





Lasso (L1) regularization

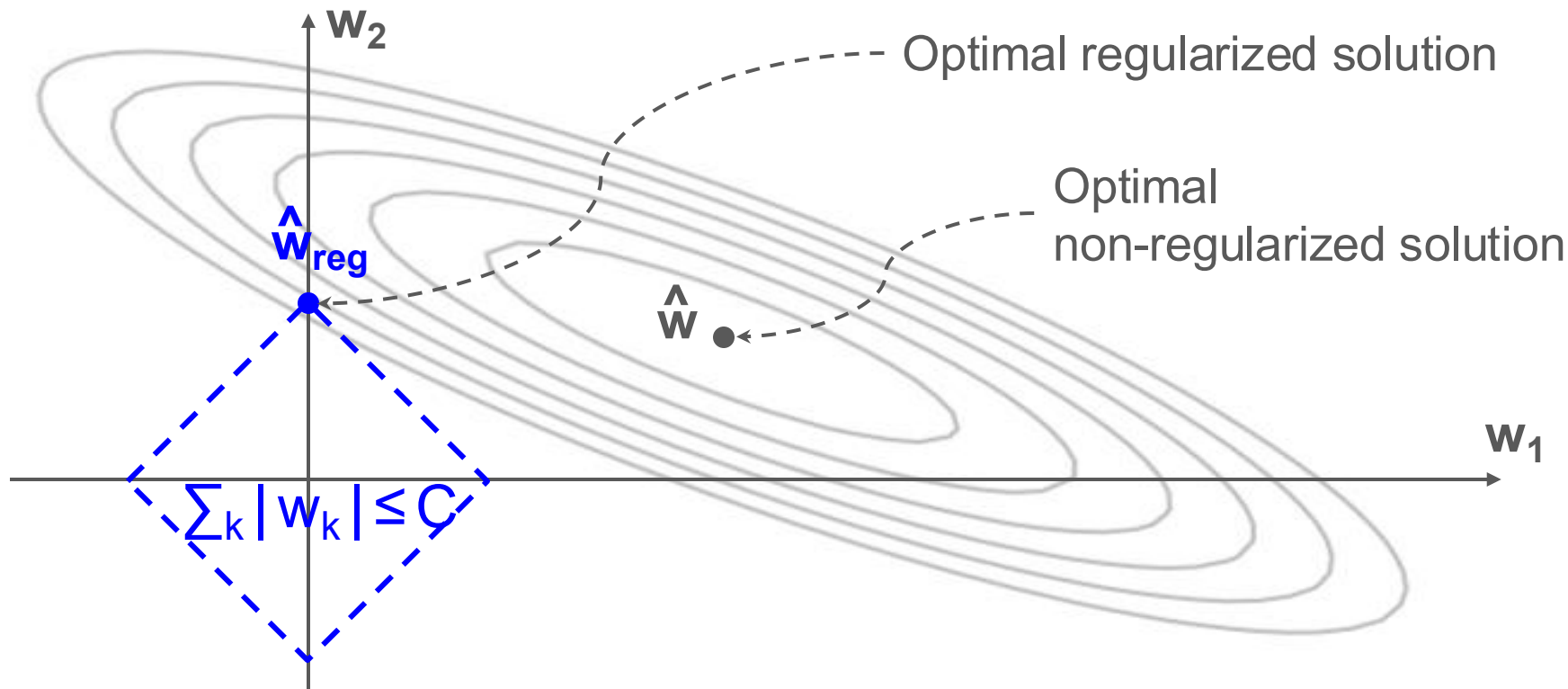
- Minimizes the following augmented error score:

$$\text{MSE}_{L_1}(\mathbf{w}) = \text{MSE}(\mathbf{w}) + \lambda \sum_k |w_k|$$

- Where $\lambda > 0$ controls the relative impact of the penalty
 - Large λ : the coefficients w_k are forced to be small
- Minimizing augmented error score MSE_{L_1} is equivalent to solving the constraint minimization problem:

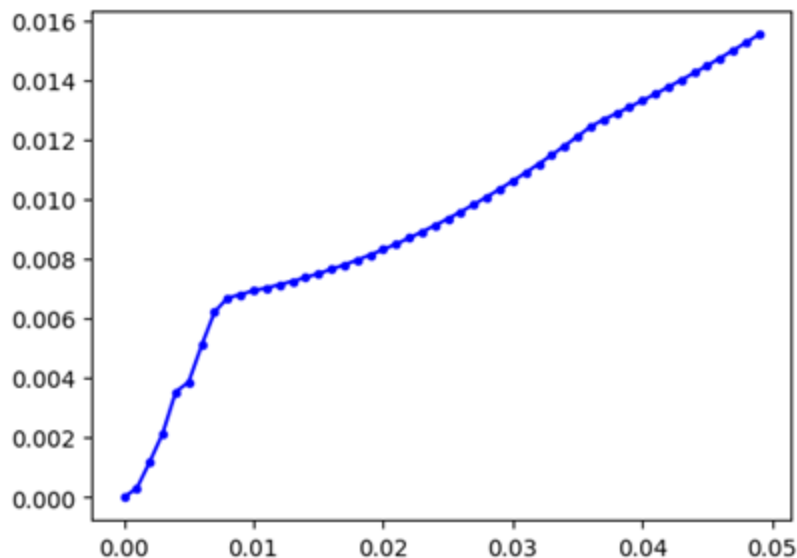
$$\text{argmin } \text{MSE}(\mathbf{w}) \text{ subject to } \sum_k |w_k| \leq C$$

Understanding the role of the constraint region (Lasso)



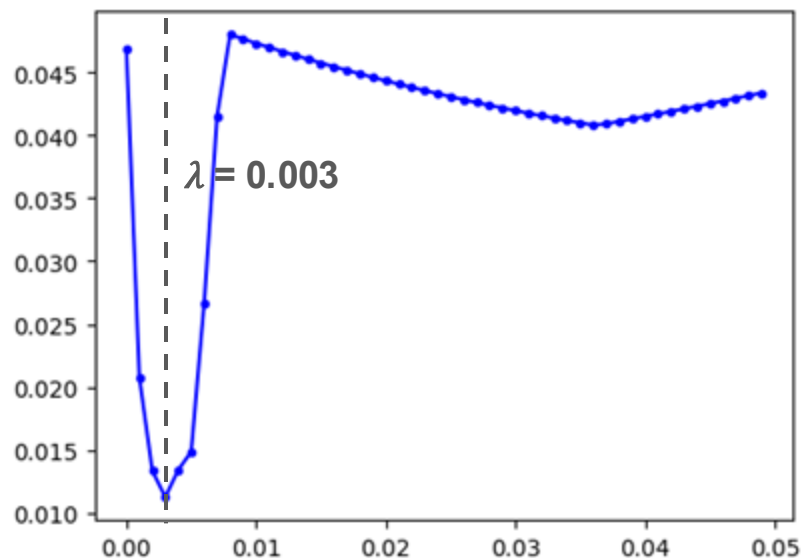
In- and out-of-sample errors as functions of λ (Lasso)

In-sample Errors



Shrinkage factor λ

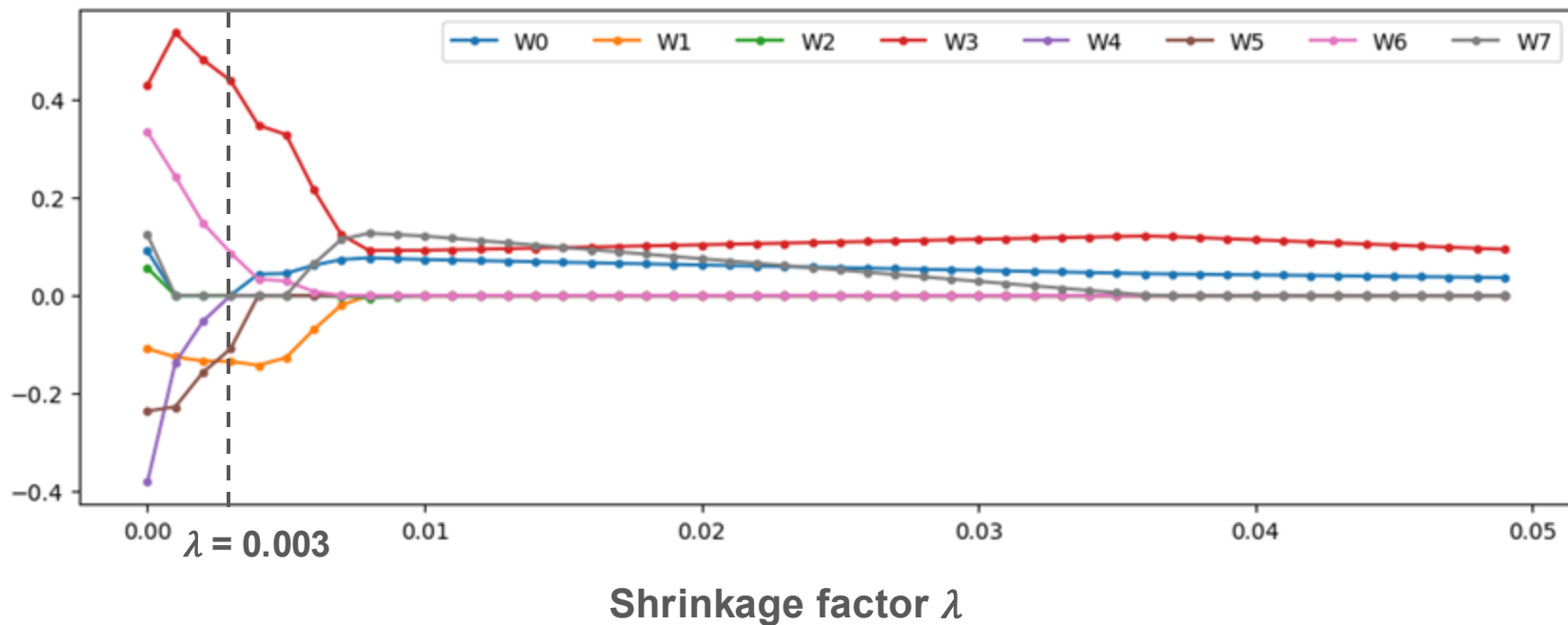
Out-of-sample Errors



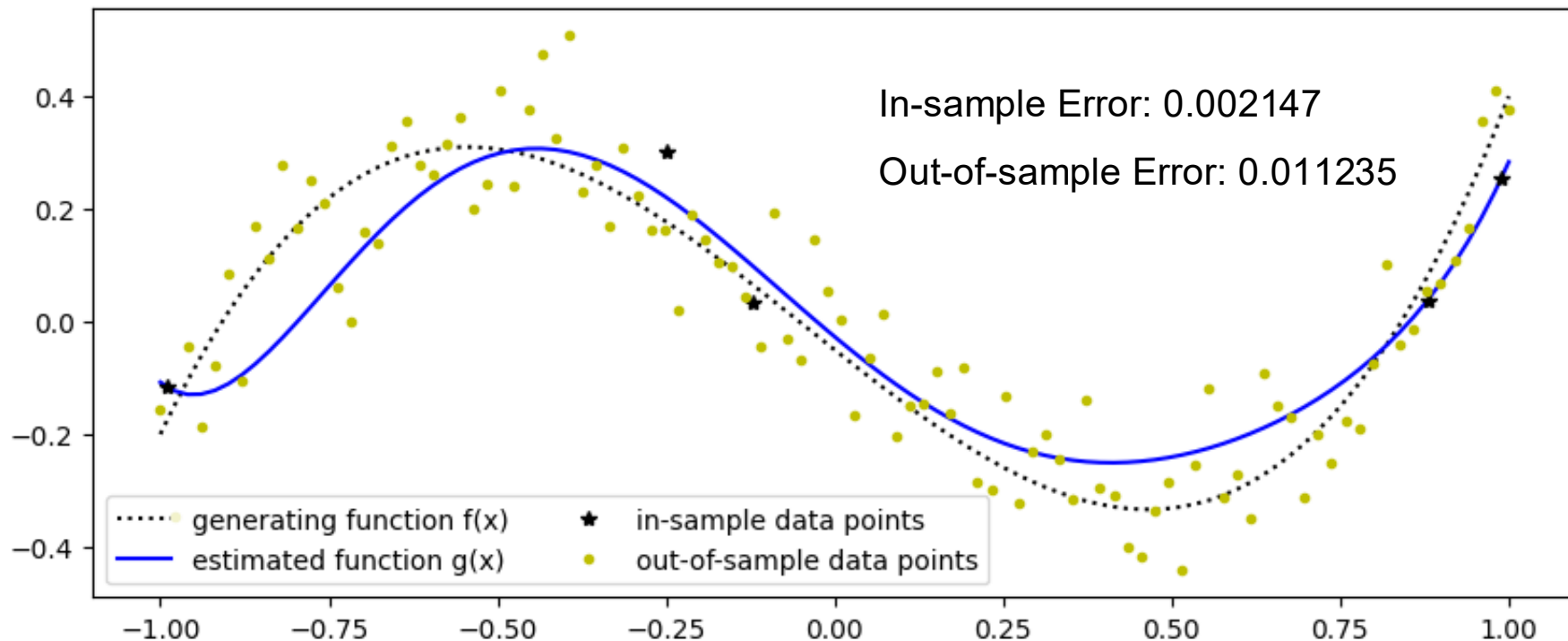
Shrinkage factor λ

Optimal model coefficients as functions of λ (Lasso)

Optimal model coefficients

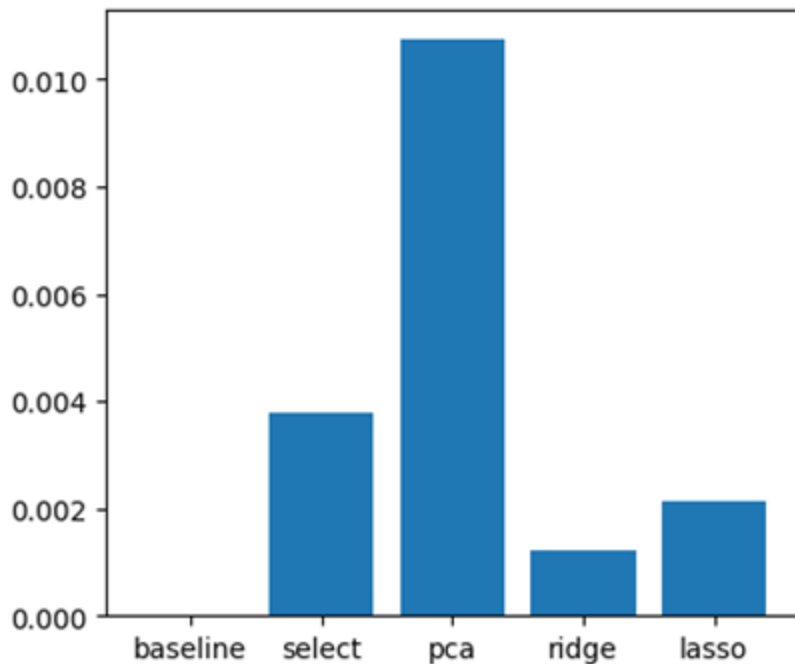


Lasso regression model (with $\lambda = 0.003$)

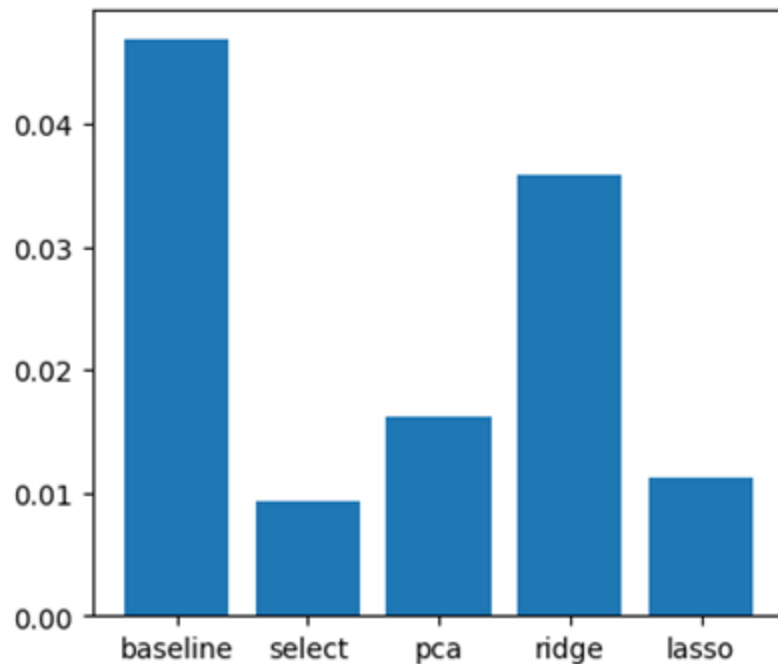


Summary of results: subset selection and regularization

In-sample Errors



Out-of-sample Errors





Some final remarks on regularization

- So far we have considered cases in which the regularization parameter λ is common to all model coefficients
- Penalty terms implementing different λ s to each coefficient w_k can also be used, for instance $\sum_k \lambda_k w_k^2$ or $\sum_k \lambda_k |w_k|$
 - $\lambda_k = k$ or $\lambda_k = e^k$ favors fitting w_k s with small k
 - $\lambda_k = (1-k)^{-1}$ or $\lambda_k = e^{-k}$ favors fitting w_k s with large k
- Subset selection can be seen as a special case of regularization, in which $\lambda_k = 0$ for the selected variables and $\lambda_k \rightarrow \infty$ otherwise



Week 4 - Session 2

Cross-validation

- Cross-validation
- K-fold cross-validation
- Leave-one-out cross-validation

Some important considerations for the ML practice

- Out-of-sample data is of critical value for success:
 - To assess whether we are learning or overfitting
 - To select good models to reduce the risk of overfitting
- However, out-of-sample data is not a free resource:
 - Data is limited and, in most of the cases, scarce
 - Splitting data into in- and out-of-sample sets poses a compromise between learning and assessing performance
- Data should be used wisely, cross-validation is a way of doing it

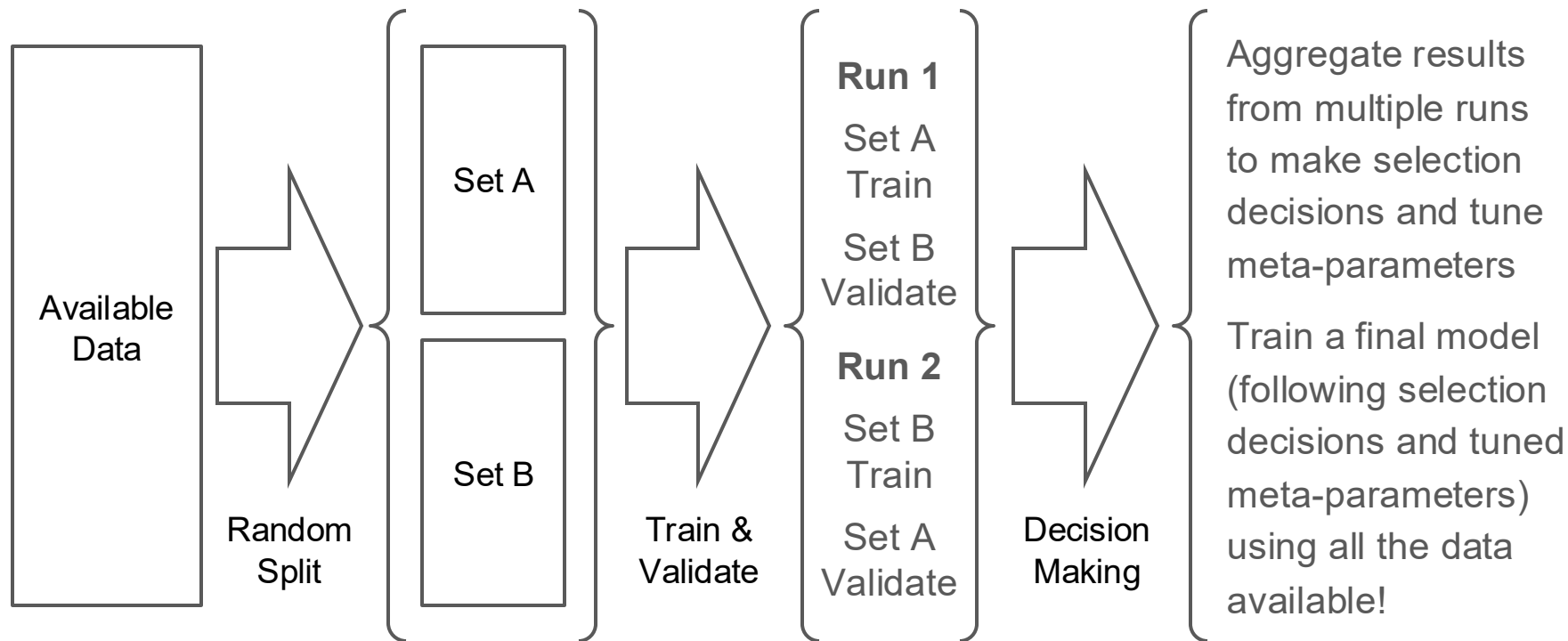
The validation set approach

- Randomly divide the available data into two sets:
 - A training dataset (50%)
 - A validation dataset (50%)
- Train models by using the train dataset
- Evaluate models/variants by using the validation dataset:
 - Assess whether overfitting might be occurring
 - Select best performing model/variant based on results

Drawback of the validation set approach

- There are two opposite forces into play:
 - The larger the train dataset the better the learning
 - The larger the evaluation dataset the better the estimation of the out-of-sample performance
- Two tricks help to overcome this fundamental limitation:
 - Cross-validation: validate across multiple subsets
 - Bootstrapping: resampling with replacement

Cross-validation: validating across multiple subsets



K-fold cross-validation: validating across K subsets

Follows the same main idea of cross-validation:

- Randomly divide the data into K sets of similar size
- Conduct K training/validation experiments for $k=1, 2, \dots, K$:
 - Use set k for validation purposes
 - Use the rest of the K-1 sets for training purposes
- Aggregate results from K training/validation experiments (mean error values the K folds) for decision making in model selection
- Train the final model by using all data available

Leave-one-out (LOO) cross-validation

Takes the K-fold cross-validation principle to the limit:

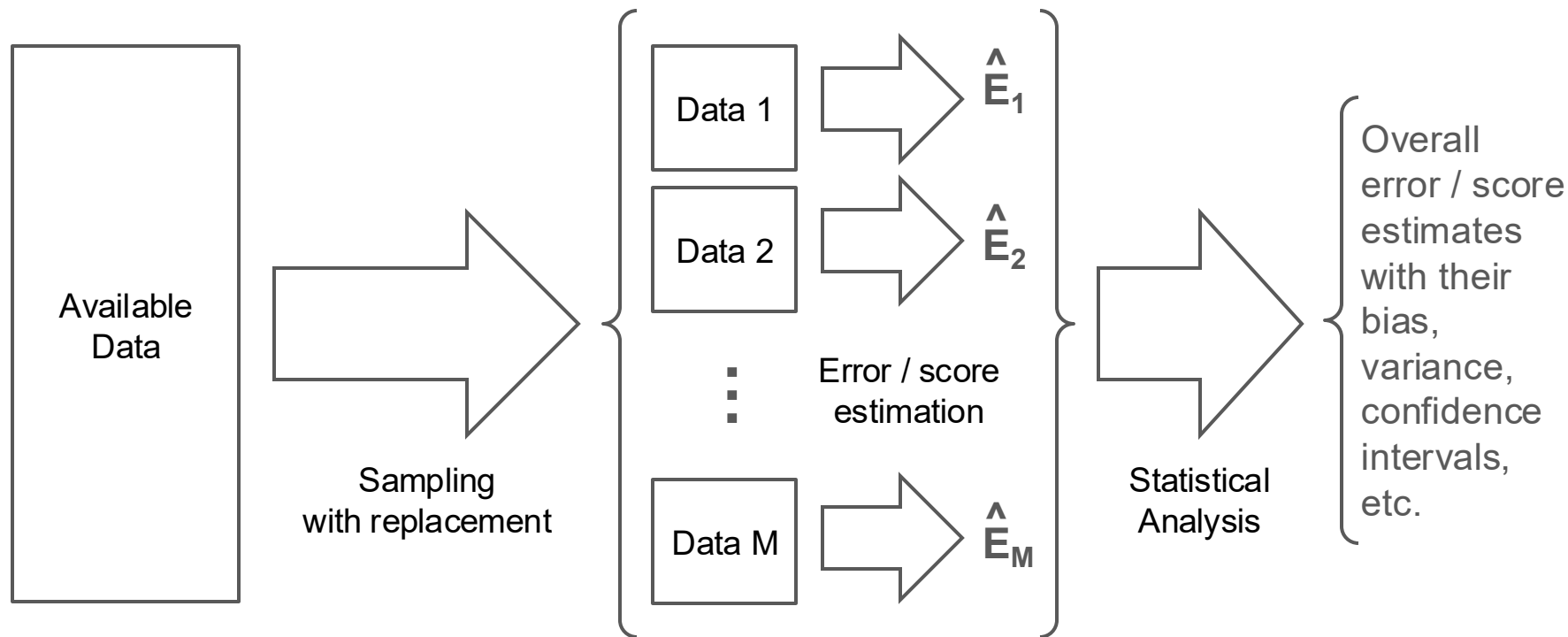
- Conduct N training/validation experiments for $k=1,2,\dots,N$, where N is the total number of data samples available:
 - Use one individual sample for validation at a time
 - Use the rest of data samples for training purposes
- Aggregate results from N training/validation experiments (mean error values the N folds) for decision making in model selection
- Train the final model by using all data available

Leave-one-out vs K-fold cross-validation vs cross-validation

	Validation	K-fold X-val	LOO X-val
Computational Cost	Small	Intermediate	High
Estimation Bias	High	Intermediate	Low
Estimation Variance	Low	Intermediate	High*
Common Setting	K = 2	K = 5 or K = 10	K = N

* Although counter intuitive, it is know that the mean of highly correlated quantities has higher variance than the mean of uncorrelated ones

Bootstrapping: resampling with replacement



Bootstrapping fundamentals

- Main assumption: inference of the true probability distribution (given the available data) is closed to an inference of an empirical distribution (given the resampled data)

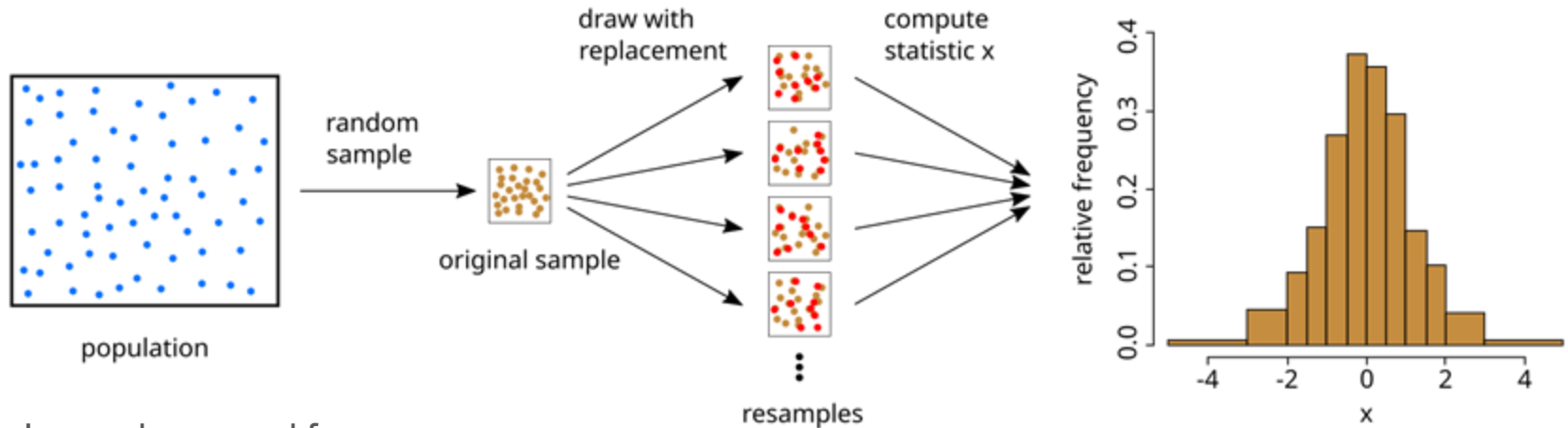


Image borrowed from:

[https://en.wikipedia.org/wiki/Bootstrapping_\(statistics\)#/media/File:Illustration_bootstrap.svg](https://en.wikipedia.org/wiki/Bootstrapping_(statistics)#/media/File:Illustration_bootstrap.svg)



A bootstrapping example

- Consider a set of targets and predictions as follows:

	0	1	2	3	4	5	6
targets	1.5	2.2	3.4	1.3	5.7	4.2	8.8
predictions	1.8	2.0	3.8	1.0	5.4	4.6	7.8

- For which the MSE can be estimated to be: 0.232857
- Use the bootstrapping technique estimate the bias, variance, and standard deviation of the reported MSE.
- Plot the empirical distribution and estimate the probability of $MSE > 0.5$



A bootstrapping example

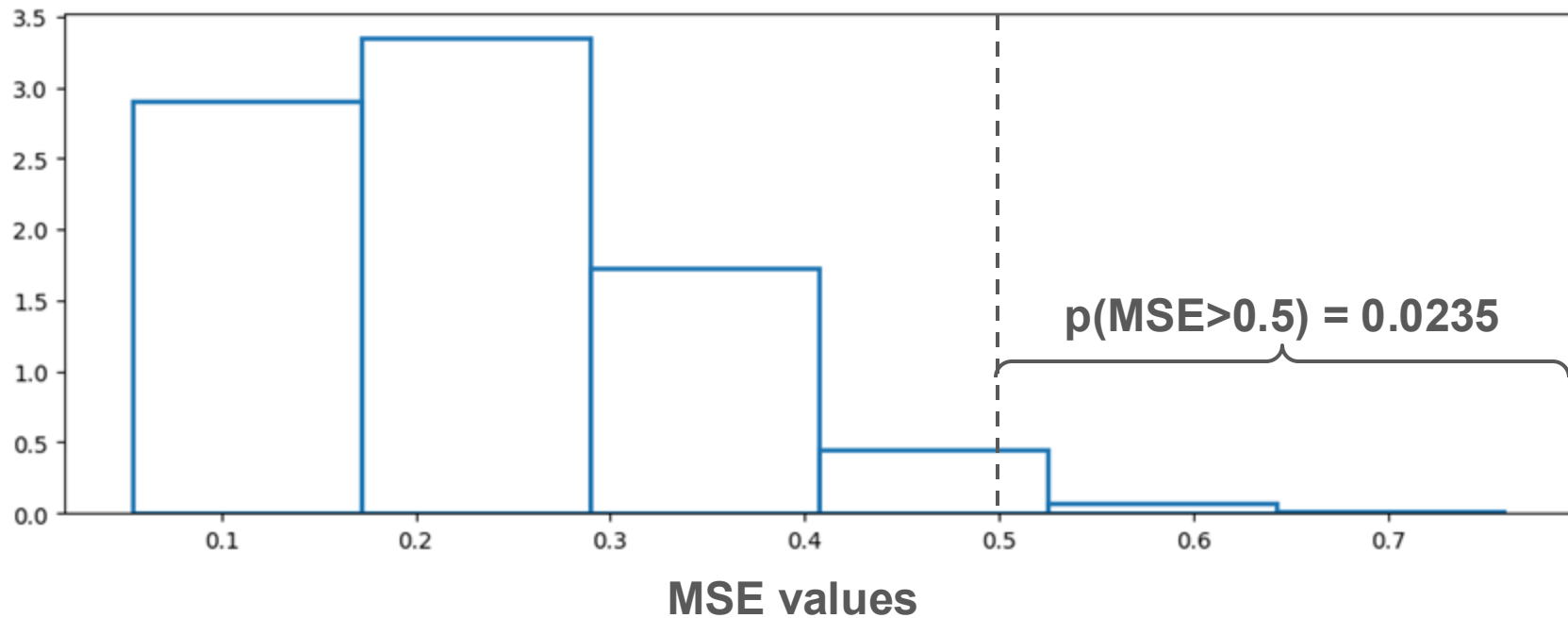
- Using sampling with replacement we generate 10000 scores:

```
tmp = []  
for k in range(10000):  
    sd = data.sample(n=data.shape[0], replace=True)  
    tmp.append(mse(sd['targets'].values, sd['predictions'].values))  
mse_bootstraps = np.array(tmp)
```

mean	variance	std
0.232498	0.014127	0.118858

- Then, we can compute statistics...
- The original estimate is unbiased:
 - Estimated MSE (0.2328) $\approx$ mean of empirical distribution (0.2324)

A bootstrapping example





Hands-on Coding Session W4S1

Hands on coding session W4S1



- Get familiar with the notebook for W4S1 and its contents
- Look at the documentation for some of the functions used there from python modules:
 - Scikit-Learn: <https://scikit-learn.org/0.16/index.html#>
- Work on exercises in Part 1 (subset selection methods)
- Work on exercises in Part 2 (regularization methods)
- Work on exercises in Part 3 (applying cross-validation)

Week 4 - Closing remarks

Wrap-up and Homework 4

- What did we look at today?
- Further reading and support materials
- Fourth homework assignment

What did we look at today?

Overfitting and Regularization	● Overfitting ● Subset Selection ● Regularization
Cross-validation	● Cross-validation ● K-fold cross-validation ● Leave-one-out cross-validation
Hands-on coding session W4S1	● Subset Selection Methods ● Regularization Methods ● Applying Cross-validation

Further reading and support materials

- James, Witten, Hastie, Tibshirani, Taylor, “An Introduction to Statistical Learning with Applications in Python” (Chapters 5 and 6) <https://www.statlearning.com/>
- Abu-Mostafa, Magdon-Ismael, Lin, “Learning From Data” (Chapter 4) <https://work.caltech.edu/telecourse>

Fourth homework assignment (due date 07/24)

- Complete all coding exercises from coding session W4S1



Term project proposal presentations 07/24

Presentations will be held after the midterm exam. Please plan for 7-min presentations plus 2-3 mins of Q&A. Your presentation must include:

- **Presentation:** your team name and members
- **Introduction:** What is your project about? What is the problem you are trying to solve? Any previous related works?
- **Motivation:** Why you want to work on such problem? What is the impact/benefit of it? Who benefits from it?
- **Work plan:** What datasets are you going to use? What types of models are you considering? What evaluation metrics are you going to use? What baseline systems are you using as reference?
- **Risk analysis:** What can go wrong in your project? Any risk mitigation strategy in mind?



Remember:

Office Hours

Tuesday 07/22 6:00-7:00pm

Next Class

Thursday 07/24, 5:30-9:30pm

Fourth Homework Due

Thursday 07/24, 11:59pm

Midterm

Thursday 07/24, 6:00-8:00pm